

possible. See the documentation for details.

You can avoid future appearances of this warning by setting the value of the 'store-plaintext-passwords' option to either 'yes' or 'no' in '/home/stylesheet/.subversion/servers'.

Store password unencrypted (yes/no)? yes

Checked out revision 0.

\$

This is really required because, in older releases of Subversion, the user is unaware of the plain text storage of passwords. The encrypted storage also applies for SSL client certificate passphrases in Subversion 1.6.

Ctypes Python bindings

Subversion 1.6 has support for ctypes Python bindings, which have many advantages over the Swig-based Python bindings. The ctypes Python bindings are generated automatically, which provides access to all Subversion APIs and new features immediately, unlike Swig bindings which must be updated by the developer team for every release to get the latest APIs working.

The ctypes bindings are pure Python and cross-platform. You can generate the function definitions on one platform and copy them to another platform without incident. In other words, users don't need a compiler to install the binding. The ctypes bindings are both forward and backward compatible with different versions of Subversion and with different libraries, as long as the functions used in the programs have compatible definitions. Besides the low-level bindings, the ctypes bindings have high-level classes, which make it easy to access common Subversion functionality in a 'Pythonic' way.

FSFS packing

Subversion 1.5 introduced the ability for FSFS repositories to be sharded into multiple directories. Subversion 1.6 takes the sharding concept further, and allows full shard directories to be *packed* into a single file. By reducing internal fragmentation in the filesystem, packed FSFS repositories have significant space savings over their unpacked counterparts, especially repositories that contain many small commits. Using a one-file-per-shard approach also allows Subversion to reduce disk I/O and better exploit operating system caches.

In order to pack a repository, we can use the 'svnadmin pack' on the repository. Once the repository is packed, there is no means by which it can be brought back to the unpacked state, which prevents serving of this repository from servers older than Subversion 1.6.

FS representation sharing

In a large repository with many branches and merges between these branches, it is common that many files will have similar history and content. In earlier versions of Subversion, this is stored as deltas against previous versions of the file, but Subversion 1.6 will use existing representation in the filesystem for duplicate storage. Though it is not an improvement visible to users, depending on the size of the repository and the degree of branching and merging, this can cause up to 20 and 15 per cent of space reduction for Berkeley DB and FSFS-based repositories, respectively.

Single file externals

Subversion 1.6 supports single file externals. If the svn:externals URL points to a single file, this gets added to the working copy as a versioned item. The file externals cannot be moved or deleted like a normal versioned file, in which case the user must change the svn:externals property, but file externals could be copied. File externals can be identified by an 'x' in the switched status column of the *svn status* command.

Command line syntax for relative URLs

Subversion 1.6 provides the user with a convenient way to refer the repository URLs, where a user need not type the entire URL of the repository, but can use a relative path inside the repository. The root of any Subversion repository can be defined as '^/', the relative path inside the repository follows it. For example, if a user wants to refer a directory sandbox inside the trunk of the repository, the following notation could be used:

```
$ svn SUBCOMMAND ^/trunk/sandbox
```

Other key features and improvements

In Subversion 1.6, there are many more new features available that are not covered in this article. They are:

- Logging support for *svnserve*
- BDB repository reverse deltas
- Support for memcache in FSFS repositories

Apart from this, there are many more improvements to the existing features and lots of bug fixes:

- svn log can take multiple revisions
- The Trust server certificate option is added
- API changes and language bindings improvements

The latest release of Subversion (1.6) source can be downloaded from subversion.tigris.org. If you want Subversion binaries for different platforms, visit open.collab.net. 

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